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DEFENCE LABORATORY, JODHPUR

A

Project Training Report

On

DESIGN AND DEVELOPMENT OF AN RO MEMBRANE SELECTION AND PERFORMANCE PREDICTION SOFTWARE FOR WATER PURIFICATION SYSTEMS

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SUBMITTED TO:

DR. VIKASH CHANDRA JANU

Sc-F

DEFENCE LABORATORY

JODHPUR, RAJASTHAN

SUBMITTED BY:

MAYTRI SINGH AHLUWALIA

B.TECH CSE (IOT)

3rd year

SKIT COLLEGE,

JAIPUR, RAJASTHAN

CERTIFICATE

This is to certify that the report on the "DESIGN AND DEVELOPMENT OF AN RO MEMBRANE SELECTION AND PERFORMANCE PREDICTION SOFTWARE FOR WATER PURIFICATION SYSTEMS" submitted by Maytri Singh Ahluwalia, a student at SKIT College, Jaipur, Rajasthan, has successfully completed her internship at Defence Laboratory (DRDO), Jodhpur in the Water Quality Management Group from 18th May 2026 to 30th June 2026. During the internship period, Maytri Singh Ahluwalia has completed her assigned task.

GUIDED BY:

DR. VIKASH CHANDRA JANU

Sc – F

DEFENCE LABORATORY, JODHPUR

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ABSTRACT

Reverse Osmosis (RO) is a widely used water purification technology for removing dissolved salts and impurities from water. The design and operation of an RO system require careful selection of membrane and accurate prediction of system performance based on feed water characteristics and operating conditions.

This project presents the development of an RO Membrane Selection and Performance Prediction Software for Water Purification Systems. The software enables users to analyse feed water quality, assess scaling and fouling risks, select suitable membranes, configure system parameters, and predict important performance indicators such as product flow rate, product TDS, feed pressure, flux, salt rejection, and osmotic pressure.

The system also includes features such as membrane comparison, project management, user authentication, and automated PDF report generation. Developed using React and Flask, the software provides a user-friendly web interface for efficient RO system design and analysis.

The accuracy of the developed software was evaluated through comparison with ROSA (Reverse Osmosis System Analysis) software. The results demonstrated that the developed system provides reliable performance predictions and serves as an effective tool for RO membrane selection and water treatment system design.

CHAPTER 1

INTRODUCTION

1.1 Background

Water is an essential natural resource required for drinking, agriculture, industrial processes, and various defence applications. However, increasing population growth, rapid industrialization, and environmental pollution have significantly reduced the availability of clean and safe water resources. Many natural water sources contain dissolved salts, organic contaminants, microorganisms, and other impurities that make the water unsuitable for direct use. Consequently, efficient water treatment technologies have become increasingly important to meet the growing demand for high-quality water.

Among the various water treatment methods available, Reverse Osmosis (RO) has emerged as one of the most effective and widely adopted technologies for desalination and purification. RO systems utilize semi-permeable membranes to remove dissolved salts, suspended solids, and other contaminants from water, producing high-quality permeate suitable for domestic, industrial, and specialized applications. Due to their high efficiency and reliability, RO systems are extensively employed in seawater desalination plants, industrial water treatment facilities, pharmaceutical industries, power generation units, and defence establishments.

The performance of an RO system depends on several factors, including feed water quality, membrane characteristics, operating pressure, recovery rate, and system configuration. Therefore, proper membrane selection and system design are critical for achieving optimum performance, minimizing operational costs, and ensuring long-term system reliability. As water treatment requirements become increasingly complex, there is a growing need for computational tools that can assist engineers in accurately designing and evaluating RO systems.

1.2 Reverse Osmosis Technology

Reverse Osmosis (RO) is a membrane-based water purification process used to remove dissolved salts, minerals, organic compounds, and other impurities from water. The process operates by applying pressure greater than the natural osmotic pressure of the feed water, forcing water molecules through a semi-permeable membrane while rejecting most dissolved contaminants.

In a typical RO system, feed water is supplied to the membrane under high pressure. The membrane allows only purified water, known as permeate or product water, to pass through while retaining dissolved salts and impurities in a concentrated stream called reject or brine. This separation process results in the production of high-quality water suitable for various domestic, industrial, and scientific applications.

The performance of an RO system is influenced by several factors, including feed water quality, membrane characteristics, operating pressure, recovery rate, temperature, and system configuration. Proper selection of membrane elements and operating conditions is essential to achieve the desired water quality, maximize recovery, minimize energy consumption, and reduce membrane fouling and scaling.

Due to its effectiveness and reliability, Reverse Osmosis technology is widely used in seawater desalination, brackish water treatment, industrial process water production, pharmaceutical manufacturing, power generation plants, and defence-related water purification systems.

1.3 Problem Statement

Designing an efficient Reverse Osmosis (RO) system requires careful evaluation of feed water quality, membrane characteristics, operating conditions, and potential scaling or fouling risks. Traditionally, engineers perform these calculations manually or use specialized commercial software tools to estimate system performance and select suitable membrane elements. These approaches can be time-consuming, require significant technical expertise, and may lead to errors if the input data or calculations are not handled correctly.

In addition, the selection of an appropriate membrane depends on multiple parameters such as feed water type, Total Dissolved Solids (TDS), recovery rate, feed flow, and water chemistry. Evaluating all these factors simultaneously can be a complex task, especially when rapid system design and performance assessment are required.

Therefore, there is a need for a user-friendly software solution that can automate membrane selection, analyse feed water characteristics, assess scaling risks, predict system performance, and generate comprehensive reports. Such a system can reduce design time, improve accuracy, and assist engineers in making informed decisions during the design and operation of RO water purification systems.

1.4 Project Overview

The project titled “Design and Development of an RO Membrane Selection and Performance Prediction Software for Water Purification Systems” aims to provide an integrated platform for the design, analysis, and evaluation of Reverse Osmosis (RO) systems. The software has been developed to assist engineers and researchers in selecting suitable membrane elements and predicting system performance based on feed water characteristics and operating conditions.

The software allows users to enter feed water data, water chemistry parameters, and system design requirements through an interactive web-based interface. Based on the provided inputs, the system performs scaling and fouling assessments, recommends appropriate membrane elements, and predicts important performance parameters such as product flow rate, product TDS, feed pressure, flux, salt rejection, and osmotic pressure.

In addition to performance prediction, the software includes modules for membrane comparison, project management, user authentication, and automated PDF report generation. Users can save and retrieve projects, compare different membrane options, and generate detailed reports for documentation and analysis purposes.

The application has been developed using React for the frontend interface, Flask for backend processing, and a database-driven architecture for project storage and management. The software is deployed as a cloud-based web application, enabling users to access the system from any location through a web browser.

The primary objective of the project is to simplify the RO system design process, reduce manual calculations, improve decision-making, and provide a reliable tool for water treatment and desalination applications.

1.5 Objectives of the Project

The primary objective of this project is to develop a software tool that simplifies the design, analysis, and evaluation of Reverse Osmosis (RO) systems for water purification applications. The specific objectives of the project are as follows:

1. To develop a user-friendly web-based platform for RO system design and performance evaluation.

2. To analyse feed water characteristics and water chemistry parameters for accurate system assessment.
3. To recommend suitable membrane elements based on feed water quality and operating conditions.
4. To predict important performance parameters such as product flow rate, product TDS, feed pressure, flux, salt rejection, and osmotic pressure.
5. To assess scaling and fouling risks that may affect membrane performance and system efficiency.
6. To provide membrane comparison functionality for evaluating different membrane options and their specifications.
7. To enable users to save, retrieve, and manage multiple RO system design projects through a secure authentication system.
8. To generate detailed PDF reports containing system inputs, performance predictions, and design recommendations.
9. To validate the performance of the developed software through comparison with industry-standard RO design methodologies and software tools.
10. To provide an efficient and reliable decision-support tool for engineers, researchers, and water treatment professionals involved in RO system design and operation.

1.6 Scope of the Project

The scope of this project is focused on the design, analysis, and performance prediction of Reverse Osmosis (RO) systems used for water purification applications. The developed software provides an integrated platform for evaluating feed water quality, selecting suitable membrane elements, configuring system parameters, and predicting the operational performance of RO systems.

The software allows users to input feed water characteristics such as Total Dissolved Solids (TDS), temperature, flow rate, recovery percentage, and detailed water chemistry parameters. Based on these inputs, the system performs scaling and fouling assessments, recommends appropriate membrane elements, and predicts key performance indicators including product flow rate, product TDS, feed pressure, flux, salt rejection, and osmotic pressure.

The project also includes additional functionalities such as membrane comparison, project management, user authentication, cloud-based accessibility, and automated

PDF report generation. These features enhance the usability of the software and support efficient documentation and analysis of RO system designs.

The current scope of the software is limited to single-stage RO system design and performance evaluation based on predefined membrane databases and engineering calculations. Advanced features such as multi-stage system optimization, real-time monitoring, machine learning-based prediction, and integration with external water quality sensors are beyond the present scope and may be considered for future development.

Overall, the project aims to provide a practical and reliable tool for engineers, researchers, and water treatment professionals involved in the design and evaluation of RO water purification systems.

1.7 Intended Users

The developed RO Membrane Selection and Performance Prediction Software is intended for a wide range of users involved in water treatment and purification system design. The software is designed to assist users in analysing feed water quality, selecting appropriate membrane elements, and evaluating the performance of Reverse Osmosis (RO) systems.

The primary intended users of the software are:

1. Water Treatment Engineers: Engineers involved in the design, operation, and optimization of RO systems can use the software for membrane selection and performance prediction.
2. Researchers and Scientists: Researchers working in the fields of water treatment, desalination, and membrane technology can utilize the software for analysis and evaluation purposes.
3. Industrial Users: Industries requiring purified water for manufacturing, processing, or utility applications can use the software to assess RO system performance and design requirements.
4. Defence and Government Organizations: Organizations involved in water purification and desalination projects can use the software as a decision-support tool for system planning and evaluation.
5. Academic Institutions and Students: The software can serve as an educational tool for understanding RO system design principles, membrane selection criteria, and water treatment processes.

By providing a simple and interactive interface, the software enables both technical and non-technical users to perform RO system analysis efficiently and accurately.

1.8 Chapter Overview

The structure of the remaining chapters of this report is organized as follows:

Chapter 2: System Overview

This chapter describes the overall architecture and workflow of the developed software. It explains the software architecture, technology stack, database design, deployment architecture, and the interaction between various modules of the application.

Chapter 3: Project Initialization

This chapter explains the user authentication system, project creation process, project information management, saved projects module, and database storage. It describes how users securely manage multiple RO design projects within the software.

Chapter 4: Feed Water Characterization

This chapter discusses the feed water input parameters and water chemistry analysis required for RO system design. It explains the significance of parameters such as water type, feed TDS, feed flow, temperature, recovery, membrane size, pH, cations, anions, and other water quality calculations performed by the software.

Chapter 5: Scaling Assessment

This chapter presents the methodology used to evaluate scaling potential within the RO system. It describes the calculation and interpretation of Langelier Saturation Index (LSI), S&DSI, silica saturation, sulfate scaling, concentration factor, and overall scaling risk assessment generated by the software.

Chapter 6: Membrane Selection and System Configuration

This chapter explains the membrane database, membrane categories, membrane comparison module, and system configuration features. It also describes parameters

such as pressure vessels, stages, membrane elements, pump efficiency, feed pressure, and configuration summary used during RO system design.

Chapter 7: Performance Prediction and Projection Results

This chapter describes the core prediction module of the software. It explains the calculation and visualization of product flow, concentrate flow, product TDS, concentrate TDS, feed pressure, recovery, salt rejection, membrane flux, osmotic pressure, system health assessment, performance graphs, water recovery distribution, and cost estimation.

Chapter 8: Project Storage and Report Generation

This chapter discusses the project storage functionality and automated PDF report generation module. It explains how users save, retrieve, manage, and export complete RO system designs for documentation and future analysis.

Chapter 9: Implementation of Performance Calculations and Validation

This chapter presents the implementation of the mathematical models and engineering calculations used within the software. It describes the calculation methodology for membrane area, temperature correction factor, osmotic pressure, feed pressure, product flow, salt rejection, product TDS, membrane flux, scaling assessment, and fouling assessment. The chapter also includes the validation of the developed software through comparison of the generated results with reference RO performance data.

Chapter 10: Conclusion, Future Scope and References

The final chapter summarizes the achievements of the project, discusses the major outcomes of the developed software, presents possible future enhancements, and lists the references and technical resources used during the design and development of the project.

CHAPTER 2

SYSTEM OVERVIEW

2.1 Introduction

The RO Membrane Selection and Performance Prediction Software is a web-based application developed to assist engineers and researchers in the design, analysis, and evaluation of Reverse Osmosis (RO) systems. The software integrates water quality analysis, membrane selection, scaling assessment, performance prediction, project management, and report generation into a single platform.

The system has been designed to automate engineering calculations that are traditionally performed manually or through commercial software tools. By providing a user-friendly interface and automated computational modules, the software reduces design time, minimizes human errors, and improves decision-making during the design and operation of RO systems.

The application follows a modular architecture in which each module performs a specific function, ranging from feed water analysis to performance prediction and report generation. The integration of these modules enables users to efficiently evaluate different design scenarios and obtain detailed system performance predictions.

2.2 Software Workflow

The software follows a sequential workflow that mirrors the actual process followed during RO system design. The user first creates a project and enters feed water characteristics and operating conditions. The software then performs water chemistry analysis, scaling assessment, membrane selection, and system configuration. Based on the provided inputs, the performance prediction engine calculates key operational parameters and generates detailed results.

The workflow of the software can be summarized as follows:

User Registration / Login

↓

Create Project

↓

Enter Project Information

↓

Enter Feed Water Data

↓

Enter Water Chemistry

↓

Scaling & Fouling Analysis

↓

Membrane Selection

↓

System Configuration

↓

Performance Prediction

↓

Projection Results

↓

Membrane Comparison

↓

PDF Generation

↓

Save Project / Load Project

2.3 Software Architecture

The software follows a three-tier architecture consisting of a frontend layer, backend layer, and database layer.

1. Frontend Layer:
   The frontend is developed using React and provides an interactive user interface for entering data, viewing results, managing projects, and generating reports.
2. Backend Layer:
   The backend is developed using Flask and is responsible for performing engineering calculations, membrane selection, performance prediction, scaling analysis, and data processing.
3. Database Layer:
   The database stores user information, project details, saved configurations, and generated results. It enables project persistence and retrieval for future analysis.

The communication between the frontend and backend is performed through RESTful API endpoints, allowing seamless exchange of data and results.

2.4 Technology Stack

The software has been developed using modern web technologies and programming frameworks.

Frontend Technologies:
• React.js
• HTML
• CSS
• JavaScript

Backend Technologies:
• Flask
• Python

Database:
• SQLite

Deployment Platforms:
• Vercel (Frontend Deployment)
• Render (Backend Deployment)

Version Control:
• Git
• GitHub

These technologies provide scalability, maintainability, and ease of deployment while ensuring reliable software performance.

2.5 Database Design

The software utilizes a database-driven approach for managing user accounts and project information. The database stores all relevant project details, allowing users to save and retrieve RO system designs at any time.

The major entities stored in the database include:

• User Information
• Project Information
• Feed Water Parameters
• Water Chemistry Data
• System Configuration Details
• Projection Results
• Selected Membrane Information

The database architecture ensures secure storage and efficient retrieval of project data while supporting multiple users and multiple project records.

2.6 Deployment Architecture

To ensure accessibility and ease of use, the software is deployed as a cloud-based web application.

The deployment architecture consists of:

Frontend Application (React)

↓

Hosted on Vercel

↓

Backend APIs (Flask)

↓

Hosted on Render

↓

Database Storage

Users can access the software through a standard web browser without requiring local installation. This deployment approach enables remote accessibility, centralized management, and simplified software maintenance.

2.7 Summary

The developed software provides an integrated platform for RO system design, membrane selection, scaling assessment, and performance prediction. Through its modular architecture, cloud-based deployment, and automated computational capabilities, the software serves as a practical engineering tool for water treatment professionals, researchers, and organizations involved in water purification system design.

CHAPTER 3

PROJECT INITIALIZATION

3.1 Introduction

Project initialization is the first stage of the RO Membrane Selection and Performance Prediction Software workflow. This module enables users to securely access the software, create new projects, manage project-related information, and store design data for future reference. By providing project management capabilities, the software ensures organized storage and efficient retrieval of RO system designs.

The Project Initialization module serves as the foundation for all subsequent operations within the software. All feed water data, water chemistry parameters, system configurations, and performance predictions are associated with a specific project, enabling users to manage multiple RO system designs simultaneously.

3.2 User Authentication

To ensure secure access and personalized project management, the software incorporates a user authentication system. Authentication allows users to create accounts, log into the system, and access only their own project records.

The authentication system consists of two primary functions:

- User Registration
- User Login

The implementation of user authentication improves data security, prevents unauthorized access, and enables efficient project management.

3.2.1 User Registration

To ensure secure access and personalized project management, the software incorporates a user authentication system. Authentication allows users to create accounts, log into the system, and access only their own project records.

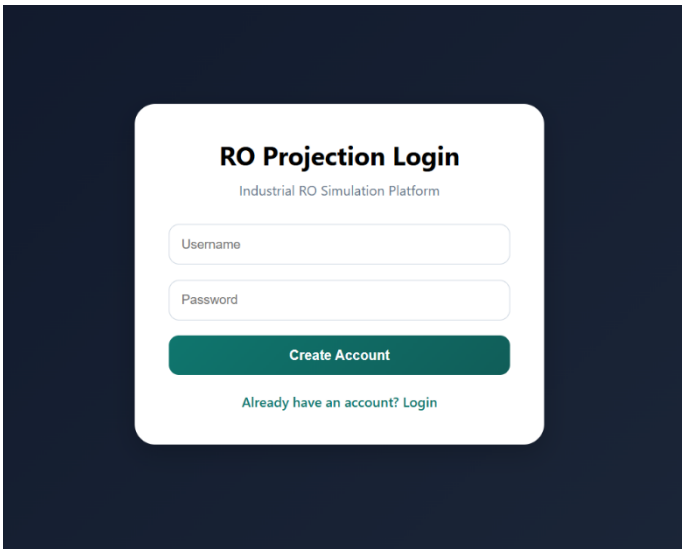


Fig 3.1 User Registration Interface

The registration process enables the creation of personalized workspaces for each user and ensures that project information remains associated with the correct account.

Benefits of User Registration:

- Secure user identification
- Personalized project management
- User-specific project storage
- Easy retrieval of previously saved projects

3.2.2 User Login

The login module allows registered users to access the software using their credentials. Upon successful authentication, users gain access to all software functionalities and project management features.

The login mechanism ensures that users can securely access their saved projects while maintaining data privacy and integrity.

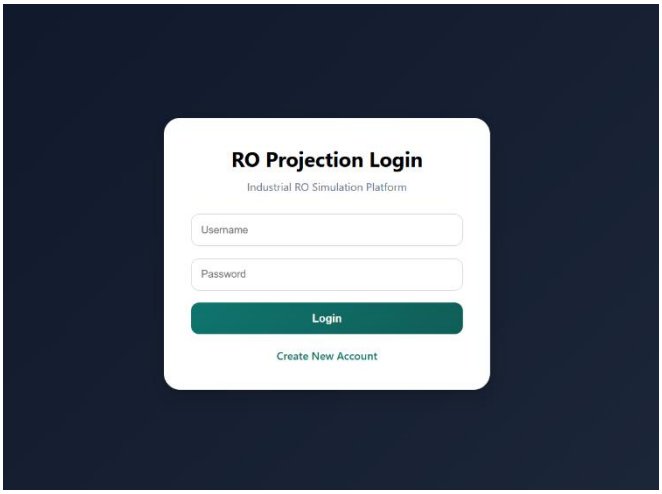


Fig 3.2 User Login Interface

Benefits of User Login:

- Secure software access
- Protection of project data
- User-specific project visibility
- Centralized project management

3.3 Project Creation

After successful authentication, users can create a new RO system design project. The project creation feature provides a structured environment for storing all design-related information.

The creation of a project serves as the starting point for subsequent activities such as feed water analysis, membrane selection, system configuration, and performance prediction.

3.4 Project Information Module

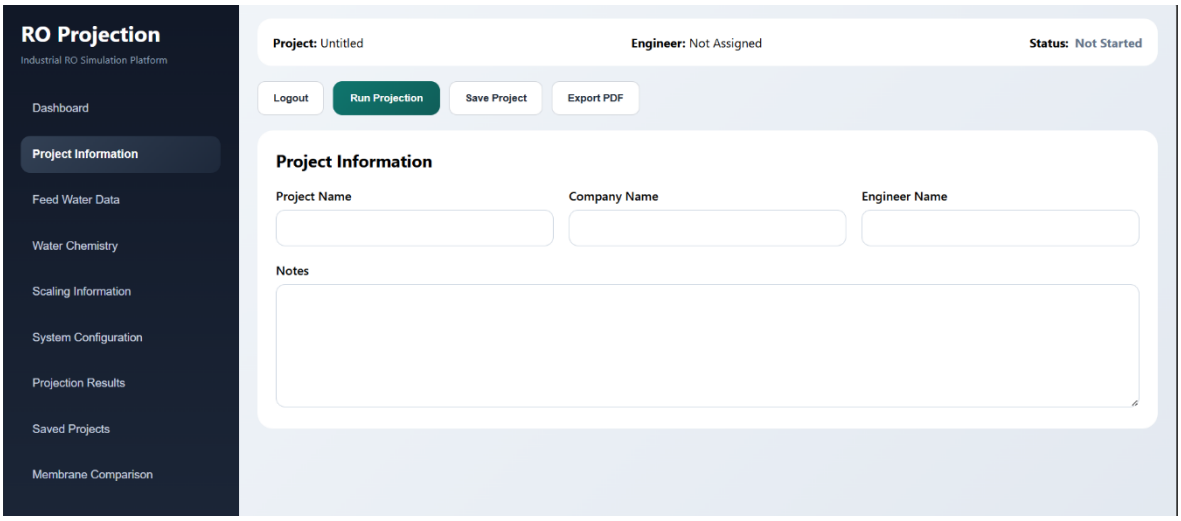


Fig 3.3 Project Information Module

The Project Information module stores administrative and identification details related to a specific RO system design project. This information helps users organize multiple projects and maintain proper documentation.

Typical project details include:

• Project Name
• Engineer Name
• Organization Name
• Project Description

The project information module improves project traceability and simplifies future retrieval and reporting.

3.5 Saved Projects Module

The software includes a Saved Projects module that enables users to store and retrieve previously created RO system designs. This functionality eliminates the need to repeatedly enter the same design parameters and allows users to continue work from previously saved states.

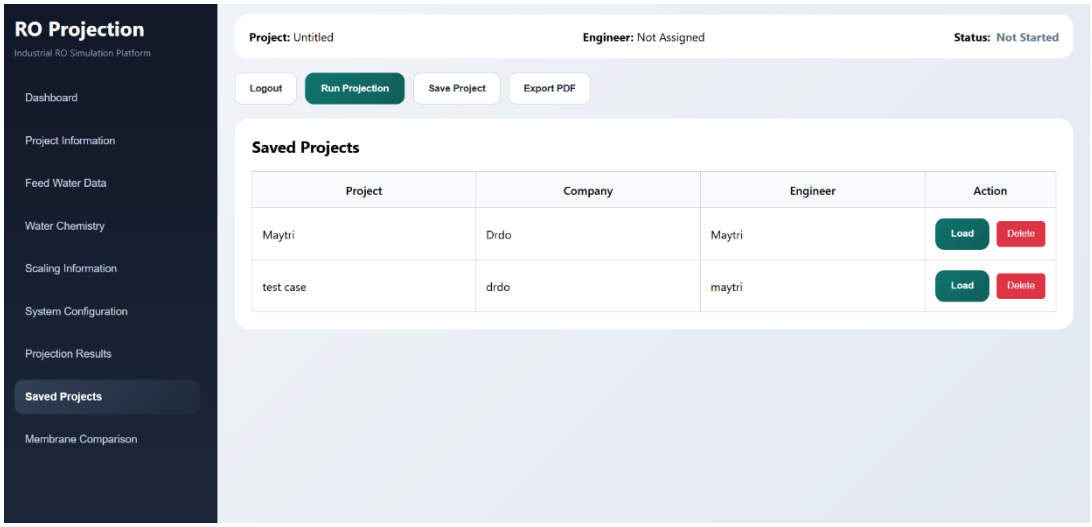


Fig 3.4 Saved Projects Management Interface

The module supports:

- Project Saving
- Project Retrieval
- Project Updating
- Project Re-analysis

This functionality significantly improves user productivity and project management efficiency.

3.6 Database Storage

All user accounts and project-related information are stored within the software database. The database maintains records of:

- User Credentials
- Project Information
- Feed Water Parameters
- Water Chemistry Data
- System Configuration Details
- Projection Results
- Selected Membrane Information

The database-driven architecture ensures secure storage, data consistency, and reliable project retrieval.

3.7 Advantages of the Project Initialization Module

The Project Initialization module provides several advantages:

- Secure user authentication
- Efficient project organization
- Long-term project storage
- Quick project retrieval
- Reduced data duplication
- Improved workflow management

These features contribute to a structured and user-friendly environment for RO system design and analysis.

3.8 Chapter Conclusion

This chapter discussed the Project Initialization module, which serves as the entry point to the software. The implementation of user authentication, project creation, project information management, and project storage establishes a secure and organized framework for conducting RO system analysis. The next chapter focuses on Feed Water Characterization, where users provide the input parameters required for RO system design and performance prediction.

CHAPTER 4

FEED WATER CHARACTERIZATION

4.1 Introduction

Feed water characterization is a critical stage in the design and analysis of Reverse Osmosis (RO) systems. The performance of an RO membrane system is directly influenced by the physical and chemical properties of the feed water. Accurate characterization of feed water enables reliable membrane selection, performance prediction, scaling assessment, and system configuration.

The Feed Water Characterization module of the developed software allows users to enter operating conditions and detailed water chemistry parameters required for RO system analysis. The collected data serves as the primary input for all subsequent calculations and performance predictions.

4.2 Feed Water Data

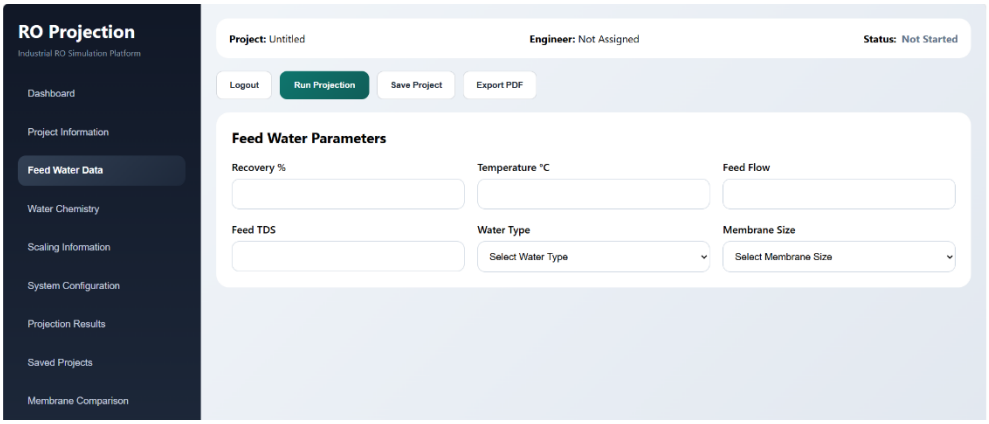


Fig 4.1: Feed Water Data Entry Interface

The Feed Water Data module collects the fundamental operating parameters required for RO system design. These parameters define the quality and quantity of water entering the system and significantly influence membrane performance.

The primary parameters considered by the software include:

- Water Type
- Feed TDS
- Feed Flow Rate
- Temperature
- Recovery Percentage
- Membrane Size

The entered values are used by the performance prediction engine to estimate product water quality, feed pressure requirements, membrane flux, and salt rejection characteristics.

4.2.1 Water Type

The software supports different categories of feed water, primarily seawater and brackish water. The classification of feed water is important because different water sources exhibit varying salinity levels, operating pressures, and membrane requirements.

Seawater generally contains very high concentrations of dissolved salts and therefore requires high-pressure membranes with superior salt rejection characteristics. Brackish water contains lower salinity levels and can be treated using membranes operating at comparatively lower pressures.

The selected water type serves as an important input for membrane recommendation and performance prediction.

4.2.2 Feed TDS

Total Dissolved Solids (TDS) represent the concentration of dissolved inorganic and organic substances present in water. TDS is commonly expressed in milligrams per litre (mg/L) or parts per million (ppm).

Feed TDS is one of the most critical design parameters because it directly affects:

• Osmotic pressure
• Membrane selection

• Operating pressure requirements
• Product water quality
• Energy consumption

As feed TDS increases, the pressure required to overcome osmotic pressure also increases. Therefore, accurate TDS measurement is essential for realistic system design and performance prediction.

4.2.3 Feed Flow Rate

Feed flow rate represents the quantity of water entering the RO system per unit time. It is generally expressed in litres per hour (LPH), cubic metres per hour (m^3/h), or gallons per minute (GPM).

Feed flow rate influences:

• System capacity
• Membrane area requirements
• Product water production
• Recovery calculations
• Pump sizing

Higher flow rates generally require larger membrane areas and larger system configurations to achieve the desired performance.

4.2.4 Temperature

Temperature affects several physical properties of water, including viscosity and diffusion characteristics. Variations in temperature can significantly influence membrane permeability and system productivity.

An increase in temperature generally results in:

• Increased water permeability
• Higher product flow rates
• Reduced water viscosity

However, excessive temperatures may negatively affect membrane lifespan and performance. Therefore, temperature correction factors are often incorporated into performance calculations.

4.2.5 Recovery Percentage

Recovery percentage is defined as the fraction of feed water converted into product water. It is one of the primary indicators of system efficiency.

Recovery (%) = (Product Flow / Feed Flow) × 100

Higher recovery values improve water utilization but may increase scaling risks due to higher concentration of dissolved salts in the reject stream.

The software utilizes recovery percentage during:

- Product flow calculations
- Concentrate calculations
- Scaling assessment
- Performance prediction

4.2.6 Membrane Size

Membrane size is a key design parameter used during system configuration and membrane selection. Different membrane sizes provide different membrane surface areas and production capacities.

The software supports membrane elements of various sizes, including:

- 2.5 × 14 inch
- 2.5 × 21 inch
- 2.5 × 40 inch
- 4 × 40 inch
- 8 × 40 inch
- 8 × 80 inch

• 14 × 40 inch
• 14 × 80 inch

Membrane size directly affects:

• Membrane surface area
• Water production capacity
• Flux calculations
• System footprint
• Number of membrane elements required

The selected membrane size is incorporated into system design calculations and performance prediction algorithms.

4.3 Water Chemistry Analysis

The Water Chemistry Analysis module is designed to evaluate the chemical composition of feed water and determine its suitability for Reverse Osmosis treatment. The module allows users to enter major cations and anions present in the feed water and automatically performs several calculations required for water quality assessment and RO system design.

The module accepts water chemistry data in terms of ion concentrations expressed in mg/L and converts them into engineering parameters such as ppm as $CaCO_3$ and milliequivalents per litre (meq/L). These calculations provide a standardized basis for evaluating water quality and performing scaling analysis**.**

4.3.1 pH

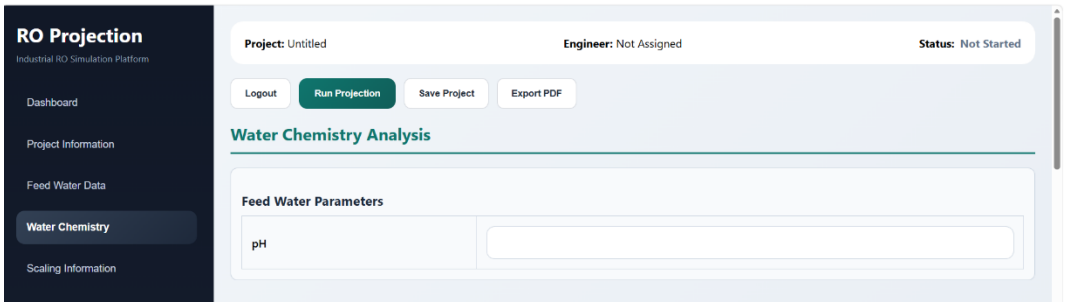


Fig 4.2: pH Analysis

pH indicates the acidity or alkalinity of water. It plays a crucial role in scaling behaviour, membrane compatibility, and chemical dosing requirements.

Variations in pH influence:

- Calcium carbonate scaling
- Silica precipitation
- Membrane stability
- Chemical treatment efficiency

4.3.2 Cation Analysis



Fig 4.3: Cation and Anion Analysis

The cation section includes the following positively charged ions:

- Ammonium (NH_4^+)
- Sodium (Na^+)
- Potassium (K^+)
- Calcium (Ca^{2+})
- Magnesium (Mg^{2+})
- Strontium (Sr^{2+})
- Barium (Ba^{2+})

These ions contribute to water hardness, salinity, ionic strength, and scaling tendencies. Calcium and magnesium are particularly important because they are major contributors to hardness-related scale formation.

4.3.3 Anion Analysis

The anion section includes:

- Chloride (Cl^-)
- Fluoride (F^-)
- Nitrate (NO_3^-)
- Carbonate (CO_3^{2-})
- Bicarbonate (HCO_3^-)
- Sulphate (SO_4^{2-})
- Silica (SiO_2)
- Boron (B)

These species influence alkalinity, salinity, scaling behaviour, and permeate water quality. Sulphate and silica concentrations are particularly important because of their tendency to form difficult-to-remove scales.

4.3.4 Water Quality Calculations

Based on the entered ion concentrations, the software automatically calculates:

- Total TDS
- Total Cations
- Total Anions
- Total Alkalinity
- Ionic Strength
- Charge Balance Error
- TDS Difference

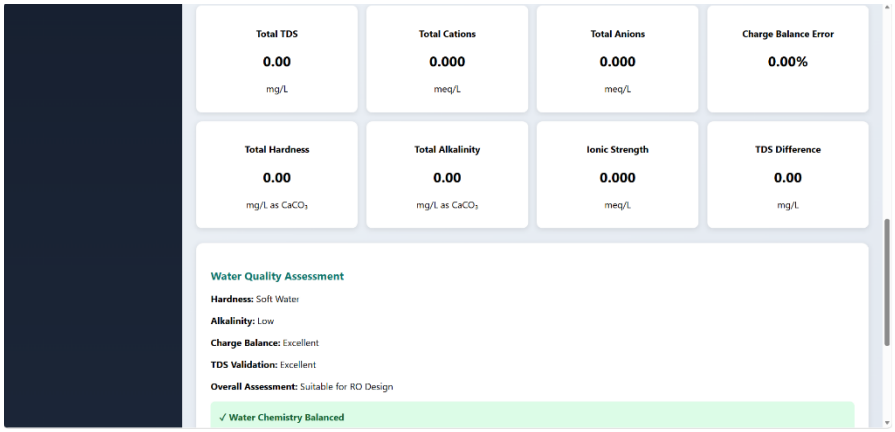


Fig 4.4: Water Quality Calculations

These parameters provide a comprehensive representation of feed water quality and are subsequently used in scaling assessment and performance prediction calculations.

4.3.5 Water Chemistry Balancing Tools

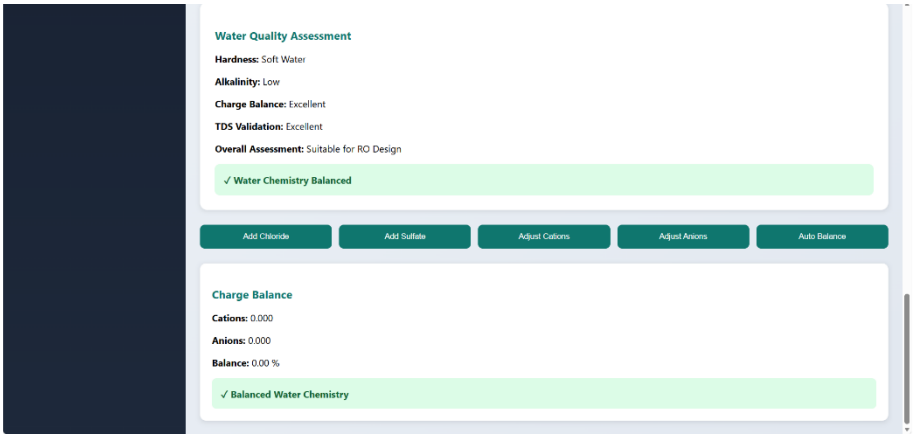


Fig 4.5: Charge Balance Verification

To improve data consistency, the software provides several balancing utilities:

- Add Chloride
- Add Sulphate
- Adjust Cations

- Adjust Anions
- Auto Balance

These tools help users correct charge imbalance and generate chemically balanced water analyses before proceeding to scaling assessment and membrane selection.

4.4 Importance of Feed Water Characterization

Accurate feed water characterization provides several benefits:

• Improved membrane selection
• Enhanced performance prediction
• Better scaling assessment
• Reduced operational risks
• Improved system reliability
• Optimized operating costs

The accuracy of software predictions depends directly on the quality of the input data provided by the user.

4.5 Chapter Conclusion

This chapter presented the Feed Water Characterization module of the developed software. The module captures both operational parameters and detailed water chemistry information required for RO system design and analysis. The collected data serves as the foundation for membrane selection, scaling assessment, system configuration, and performance prediction. The next chapter discusses the Scaling and Fouling Assessment module, which utilizes these inputs to evaluate potential operational risks within the RO system.

CHAPTER 5

SCALING ASSESSMENT

5.1 Introduction

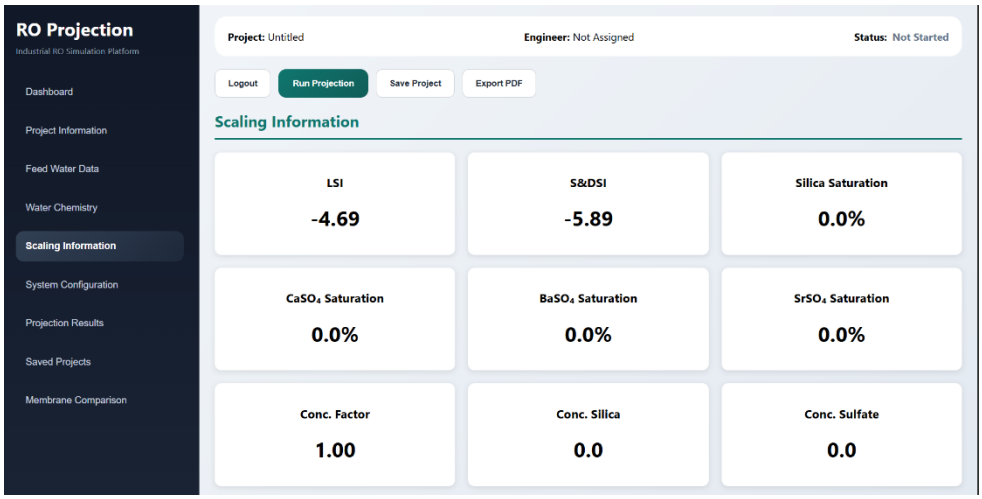


Fig 5.1: Scaling Information Dashboard

Scaling is one of the most significant operational challenges in Reverse Osmosis (RO) systems. It occurs when dissolved minerals present in feed water exceed their solubility limits and precipitate onto membrane surfaces. Scale formation reduces membrane productivity, increases pressure requirements, decreases salt rejection efficiency, and shortens membrane lifespan.

To prevent these issues, the developed software incorporates a Scaling Assessment module that evaluates scaling tendencies using feed water quality, water chemistry parameters, recovery conditions, and concentration effects. The module calculates multiple scaling indices and saturation levels to assess the potential risk of mineral precipitation within the RO system.

5.2 Purpose of Scaling Assessment

The primary objectives of the scaling assessment module are:

- Evaluate the scaling potential of feed water.
- Identify mineral precipitation risks.

- Assist in membrane selection.
- Support system configuration decisions.
- Improve operational reliability.
- Reduce membrane fouling and maintenance requirements.

The scaling assessment is performed automatically using the water chemistry information entered by the user.

5.3 Langelier Saturation Index (LSI)

The Langelier Saturation Index (LSI) is one of the most commonly used indicators for evaluating calcium carbonate scaling tendencies in water treatment systems.

The software calculates the LSI value using feed water chemistry and operating conditions.

Interpretation of LSI values:

- LSI < 0: No calcium carbonate scaling tendency.
- LSI = 0: Equilibrium condition.
- LSI > 0: Potential calcium carbonate scaling.

The calculated LSI value is displayed within the Scaling Information module and is used as a primary indicator of carbonate scale formation risk.

5.4 Sulfate and Desalination Saturation Index (S&DSI)

The Sulfate and Desalination Saturation Index (S&DSI) is used to evaluate sulfate-based scaling tendencies that may occur during RO operation.

This index considers the concentration of sulfate ions and concentration effects occurring within the membrane system.

Interpretation:

- Negative values indicate low sulfate scaling risk.
- Values approaching zero indicate increasing scaling potential.
- Positive values indicate a higher likelihood of sulfate precipitation.

The software automatically determines the sulfate scaling condition and displays the result in the scaling assessment panel.

5.5 Silica Saturation Assessment

Silica scaling is a major concern in RO systems because silica deposits are difficult to remove and can permanently affect membrane performance.

The software calculates silica saturation levels using the feed water silica concentration and concentration factor.

The generated saturation percentage provides an indication of silica precipitation risk and assists in determining safe operating conditions.

5.6 Sulfate Scaling Assessment

The software evaluates the saturation levels of the following sulfate-based scales:

5.6.1 Calcium Sulfate ($CaSO_4$)

Calcium sulfate scale may form when calcium and sulfate concentrations exceed their solubility limits.

The software calculates $CaSO_4$ saturation percentage to determine the likelihood of gypsum scale formation.

5.6.2 Barium Sulfate ($BaSO_4$)

Barium sulfate is a highly insoluble compound that can cause severe membrane scaling even at relatively low concentrations.

The software calculates $BaSO_4$ saturation levels and continuously evaluates scaling risk.

5.6.3 Strontium Sulfate ($SrSO_4$)

Strontium sulfate scaling may occur in waters containing elevated strontium concentrations.

The software evaluates $SrSO_4$ saturation percentage to determine precipitation potential.

5.7 Concentration Factor

During RO operation, dissolved salts become concentrated within the reject stream. The software calculates a concentration factor to represent this increase in dissolved constituent concentration.

The concentration factor is used in:

- Scaling calculations
- Saturation calculations
- Sulfate precipitation assessment
- Silica saturation assessment

Higher concentration factors generally increase scaling potential.

5.8 Concentrated Silica and Sulfate Analysis

To evaluate scaling under actual operating conditions, the software calculates:

- Concentrated Silica Concentration
- Concentrated Sulfate Concentration

These values represent the expected concentrations within the membrane system after concentration effects are considered.

The calculated values improve the accuracy of scaling predictions and risk assessments.

5.9 Scaling Assessment Results



Fig 5.2: Scaling Assessment Results and Risk Evaluation Panel

After completing all calculations, the software generates a comprehensive scaling assessment report.

The report includes:

- LSI Evaluation
- S&DSI Evaluation
- Silica Assessment
- Calcium Sulfate Assessment
- Barium Sulfate Assessment
- Strontium Sulfate Assessment

The results are presented in a user-friendly format to assist engineers in determining the suitability of feed water for RO treatment.

5.10 Advantages of the Scaling Assessment Module

The Scaling Assessment module provides:

- Early identification of scaling risks.
- Improved membrane protection.
- Better system reliability.
- Reduced maintenance requirements.
- Enhanced operational efficiency.
- Improved membrane lifespan.

5.11 Chapter Conclusion

This chapter presented the Scaling Assessment module of the developed software. The module evaluates multiple scaling indicators including LSI, S&DSI, silica saturation, calcium sulfate saturation, barium sulfate saturation, and strontium sulfate saturation. By analysing feed water chemistry and concentration effects, the software provides a comprehensive assessment of scaling risks and supports informed decision-making during RO system design and operation.

CHAPTER 6

MEMBRANE SELECTION AND SYSTEM CONFIGURATION

6.1 Introduction

Membrane selection is one of the most critical stages in Reverse Osmosis system design. The selected membrane determines the product water quality, recovery efficiency, operating pressure, energy consumption, and overall system performance. Different membrane types are designed for specific applications such as seawater desalination, brackish water treatment, municipal water purification, industrial water treatment, and wastewater reclamation.

The developed software contains an integrated membrane database and a system configuration module that enables users to select suitable membrane elements and configure RO system parameters. Based on feed water characteristics, scaling assessment results, and operating conditions, the software assists engineers in selecting appropriate membranes and designing optimized RO systems.

6.2 Membrane Database



Name	Type	Size	Recovery	Pressure	Rejection	Life	Application
SW30-2514	Sea Water	2.5 Inch	35%	69 bar	99.6%	3-5 Years	Small SWRO Systems
SW30-2521	Sea Water	2.5 Inch	35%	69 bar	99.6%	3-5 Years	Small SWRO Systems
SW30-4040	Sea Water	4 Inch	35%	69 bar	99.7%	3-5 Years	Sea Water Desalination
SW30HRLE-440	Sea Water	8 Inch	45%	69 bar	99.8%	3-5 Years	Sea Water Desalination
SW30-1480	Sea Water	14 Inch	45%	69 bar	99.8%	4-5 Years	Large SWRO Systems
SW30-2540	Sea Water	2.5 Inch	35%	69 bar	99.8%	3-5 Years	Commercial SWRO
BW30XFR-400	Brackish Water	8 Inch	85%	41 bar	99.7%	3-5 Years	Industrial RO
BW30PRO-440	Brackish Water	8 Inch	86%	42 bar	99.8%	4-5 Years	Municipal RO
SW30HRLE-400	Sea Water	8 Inch	42%	69 bar	99.8%	3-5 Years	Sea Water Desalination
SW30XLE-440	Sea Water	8 Inch	50%	70 bar	99.85%	4-6 Years	Marine RO
LP21-4040	Low Pressure	4 Inch	80%	20 bar	99.2%	2-4 Years	Commercial RO
ULP31-4040	Ultra Low Pressure	4 Inch	82%	18 bar	98.5%	2-4 Years	Low Energy Systems
XLE-4040	Extra Low Energy	4 Inch	85%	21 bar	99%	3-5 Years	Municipal Water
CPA5-LD	Anti Fouling	8 Inch	75%	30 bar	99.7%	4-6 Years	Wastewater Treatment

Fig 6.1: Membrane Comparison Module

The software contains a comprehensive membrane database consisting of multiple membrane models commonly used in industrial and desalination applications.

For every membrane, the database stores:

- Membrane name
- Membrane category
- Element size
- Membrane area
- Recovery capability
- Salt rejection efficiency
- Maximum operating pressure
- Expected membrane life
- Application area

The membrane database acts as the primary source of information during membrane selection and comparison.

6.3 Membrane Categories Available in the Software

6.3.1 Seawater Reverse Osmosis (SWRO) Membranes

SWRO membranes are designed to treat high-salinity seawater feed streams. These membranes operate under high pressure conditions and provide very high salt rejection performance.

Examples included in the software:

- SW30-2514
- SW30-2521
- SW30-4040
- SW30HRLE-400
- SW30HRLE-440
- SW30-1480

Characteristics:

- Pressure up to 69–70 bar
- Salt rejection up to 99.85%

- Suitable for desalination plants
- High resistance to osmotic pressure

Applications:

- Seawater desalination
- Marine water purification
- Coastal treatment facilities

6.3.2 Brackish Water Membranes

Brackish water membranes are designed for water sources having moderate dissolved salt concentrations.

Examples:

- BW30XFR-400
- BW30PRO-440

Characteristics:

- Lower operating pressure
- Higher recovery rates
- Lower energy consumption

Applications:

- Industrial RO plants
- Municipal water treatment
- Groundwater treatment

6.3.3 Low Pressure and Energy Efficient Membranes

The software also contains low-pressure membrane options such as:

- LP21-4040
- ULP31-4040
- XLE-4040

These membranes are optimized for low-energy operation while maintaining acceptable salt rejection performance.

Applications:

- Commercial RO systems
- Municipal water treatment
- Low-energy desalination systems

6.3.4 Anti-Fouling Membranes

Specialized membranes such as CPA5-LD are included for feed waters with high fouling potential.

Characteristics:

- Improved fouling resistance
- Enhanced operational stability
- Extended cleaning intervals

Applications:

- Wastewater reclamation
- Industrial effluent treatment

6.4 Membrane Performance Parameters

Recovery Rate

Recovery represents the percentage of feed water converted into product water.

Higher recovery values improve water utilization but may increase scaling tendencies.

The software stores membrane-specific recovery limits to ensure realistic system design.

Salt Rejection

Salt rejection indicates the percentage of dissolved salts removed by the membrane.

High rejection values produce low-TDS permeate water.

The software uses rejection values during product TDS prediction.

Operating Pressure

Operating pressure represents the pressure required for membrane operation.

Pressure requirements vary according to:

- Water type
- Membrane category
- Feed salinity

Seawater membranes require substantially higher operating pressures than brackish water membranes.

Membrane Life

Membrane lifespan is an important economic parameter included in the software database.

Typical values stored:

- 2–4 Years
- 3–5 Years
- 4–6 Years

This information assists engineers during membrane comparison and cost evaluation.

6.5 System Configuration Module

After selecting a suitable membrane, the user configures the RO system through the System Configuration page.

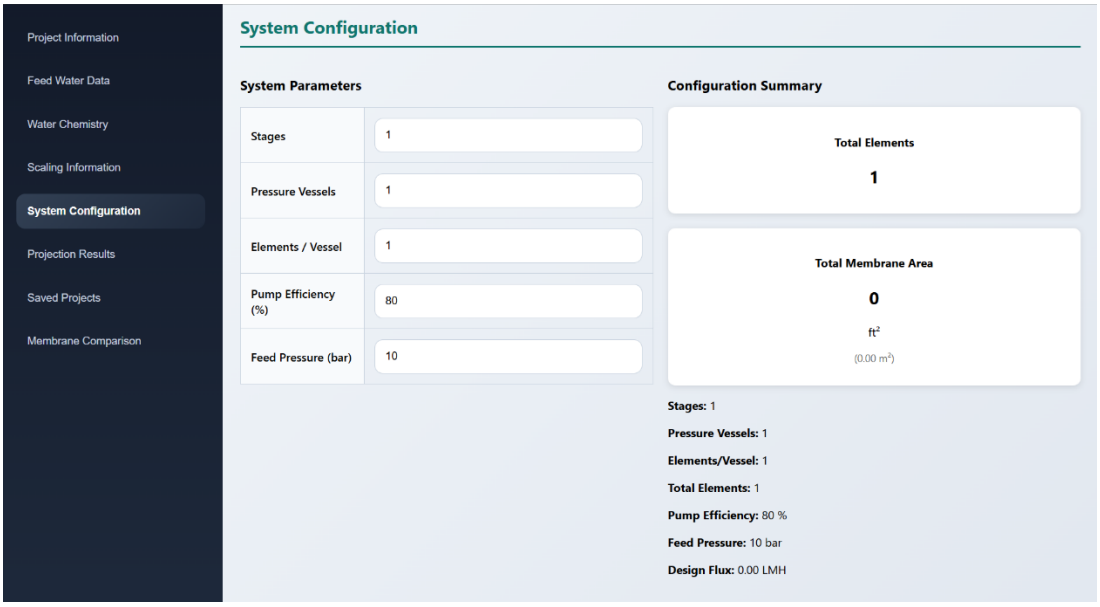


Fig 6.2 System Configuration Interface

The module allows users to define the physical layout and operational characteristics of the RO plant.

The following parameters are configurable:

- Number of stages
- Number of pressure vessels
- Elements per vessel
- Pump efficiency
- Feed pressure

6.6 Number of Stages

A stage is a group of pressure vessels operating under similar hydraulic conditions.

Multi-stage configurations are often used to:

- Improve recovery
- Optimize concentrate handling
- Reduce energy consumption

The software allows users to specify the required number of stages according to plant design requirements.

6.7 Pressure Vessels

Pressure vessels are cylindrical structures that contain membrane elements.

The number of pressure vessels directly affects:

- System capacity
- Flow distribution
- Hydraulic performance

Increasing the number of vessels generally increases system production capacity.

6.8 Elements per Vessel

Each pressure vessel contains one or more membrane elements.

The software calculates:

Total Elements = Pressure Vessels × Elements per Vessel

This value is automatically displayed within the configuration summary section.

6.9 Pump Efficiency

Pump efficiency represents the effectiveness of the high-pressure pumping system.

Pump efficiency affects:

- Energy consumption
- Operating cost

- Overall plant efficiency

Higher efficiency values result in improved energy utilization and lower operating expenses.

6.10 Feed Pressure

Feed pressure is the driving force required to transport water through the membrane surface.

The software utilizes feed pressure during:

- Product flow calculations
- Flux calculations
- Pressure drop estimations
- Performance prediction

Proper pressure selection is critical for maintaining system productivity.

6.11 Configuration Summary

The software automatically generates a configuration summary based on user inputs.

The summary includes:

- Total Elements
- Total Membrane Area
- Number of Stages
- Pressure Vessels
- Elements per Vessel
- Pump Efficiency
- Feed Pressure
- Design Flux

This feature allows engineers to verify system design parameters before executing projection calculations.

6.12 Role of Membrane Selection and Configuration in RO Design

The membrane selection and system configuration modules act as a bridge between feed water analysis and performance prediction.

These modules ensure that:

- Suitable membranes are selected.
- Scaling limitations are considered.
- System dimensions are properly defined.
- Performance predictions remain realistic.

The quality of projection results depends heavily on the accuracy of membrane selection and system configuration parameters.

6.13 Chapter Conclusion

This chapter presented the membrane database and system configuration capabilities of the developed software. The membrane comparison module enables engineers to evaluate multiple membrane options based on pressure, recovery, rejection, membrane life, and application area. The system configuration module facilitates the design of customized RO systems through specification of stages, vessels, membrane elements, pump efficiency, and operating pressure. These parameters form the foundation for the performance prediction engine discussed in the next chapter.

CHAPTER 7

PERFORMANCE PREDICTION AND PROJECTION RESULTS

7.1 Introduction

The Performance Prediction and Projection Results module is the core computational component of the developed RO Membrane Selection and Performance Prediction Software. After feed water characterization, scaling assessment, membrane selection, and system configuration have been completed, the software performs a comprehensive simulation of the RO system and predicts its operational performance.

The module calculates key performance indicators such as permeate flow, concentrate flow, recovery percentage, permeate quality, concentrate quality, flux, osmotic pressure, feed pressure, and salt rejection. In addition, the software generates performance charts, design validation reports, process flow visualization, and operational cost estimates.

The generated results assist engineers in evaluating the suitability, efficiency, and economic feasibility of the designed RO system.

7.2 Dashboard Overview

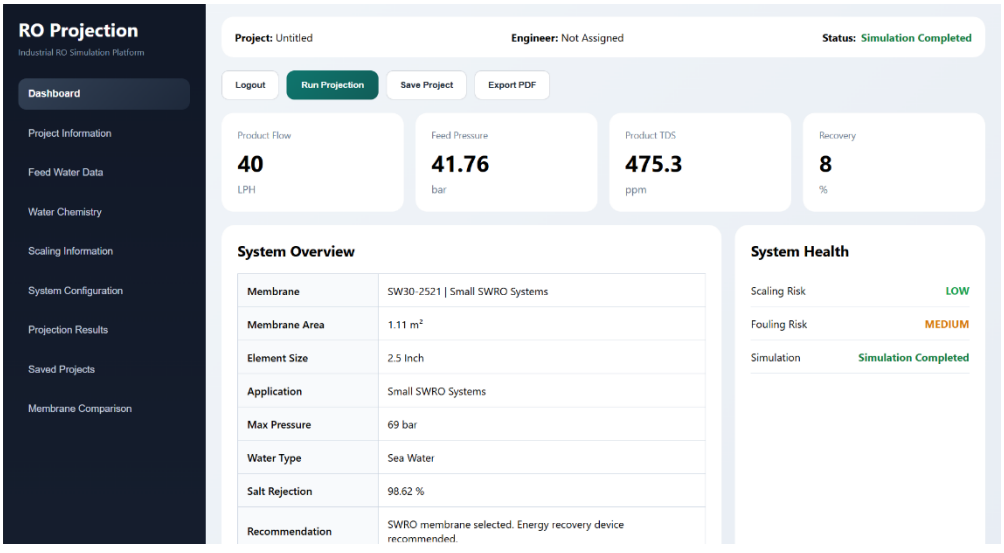


Fig 7.1: Dashboard Interface

The dashboard provides a summarized view of the most important performance parameters calculated during simulation. These parameters allow users to quickly assess the operational status of the designed RO system.

The dashboard displays:

- Product Flow
- Feed Pressure
- Product TDS
- Recovery Percentage
- System Overview
- System Health Assessment
- Membrane Information
- Scaling Risk
- Fouling Risk
- Simulation Status

These indicators provide a quick overview of system performance before detailed analysis.

7.3 Product Flow Prediction

Product flow, also known as permeate flow, represents the quantity of purified water produced by the RO system.

The software predicts permeate production using feed flow, membrane recovery, membrane area, and operating conditions.

Product flow is one of the most important design parameters because it determines the production capacity of the system.

Higher product flow values indicate increased water production capability.

7.4 Concentrate Flow Prediction

Concentrate flow represents the portion of feed water rejected by the membrane system.

The software automatically calculates concentrate flow based on feed flow and recovery percentage.

The concentrate stream contains dissolved salts and impurities removed from the feed water during treatment.

Proper concentrate management is essential for efficient RO operation.

7.5 Recovery Analysis

Recovery is defined as the percentage of feed water converted into permeate water.

Recovery (%) = (Permeate Flow / Feed Flow) × 100

The software continuously evaluates recovery performance and classifies the operating condition accordingly.

Recovery influences:

- Water utilization efficiency
- Scaling tendency
- Membrane performance
- System economics

The software displays recovery status and generates recommendations when recovery values fall outside recommended operating limits.

7.6 Product Water Quality Prediction

The software predicts the quality of permeate water by calculating permeate TDS after membrane separation.

Permeate quality is directly influenced by:

- Feed water salinity
- Membrane salt rejection
- Operating pressure
- Membrane type

Lower permeate TDS values indicate better water quality and higher membrane effectiveness.

The software automatically classifies the predicted permeate quality and displays quality indicators within the results panel.

7.7 Concentrate Water Quality Prediction

The software also predicts concentrate TDS levels generated during operation.

As recovery increases, dissolved salts become concentrated within the reject stream.

The concentrate TDS prediction assists engineers in:

- Evaluating scaling risks
- Assessing concentrate disposal requirements
- Determining operating limitations

7.8 Salt Rejection Analysis

Salt rejection is one of the primary indicators of membrane performance.

Salt Rejection (%) = ((Feed TDS − Product TDS) / Feed TDS) × 100

The software automatically calculates salt rejection using feed water and permeate quality values.

Higher rejection percentages indicate superior membrane performance and improved product water quality.

7.9 Flux Calculation

Flux represents the quantity of permeate produced per unit membrane area.

Flux is expressed in LMH (Litres per Square Meter per Hour).

The software calculates flux using:

- Product flow
- Membrane area

Flux is an important design parameter because excessive flux may accelerate fouling and scaling, while extremely low flux values may indicate oversized membrane systems.

The software performs automatic flux validation and generates recommendations accordingly.

7.10 Osmotic Pressure Prediction

Osmotic pressure represents the natural pressure exerted by dissolved salts within feed water.

The software estimates osmotic pressure using feed water salinity and ionic composition.

Osmotic pressure influences:

- Required operating pressure
- Membrane selection
- Energy consumption

The calculated osmotic pressure is used during feed pressure estimation and performance evaluation.

7.11 Feed Pressure Estimation

Feed pressure is the driving force required to overcome osmotic pressure and force water through the membrane surface.

The software predicts feed pressure based on:

- Feed water salinity
- Water type

- Membrane characteristics
- Recovery conditions

The calculated feed pressure assists engineers in selecting appropriate pumps and operating conditions.

7.12 System Health Assessment

The software performs a comprehensive health assessment of the designed RO system.

The assessment evaluates:

- Scaling Risk
- Fouling Risk
- Simulation Status
- Operating Suitability

The generated status indicators provide a quick understanding of overall system condition.

7.13 Performance Visualization

The software generates graphical representations of predicted system performance.

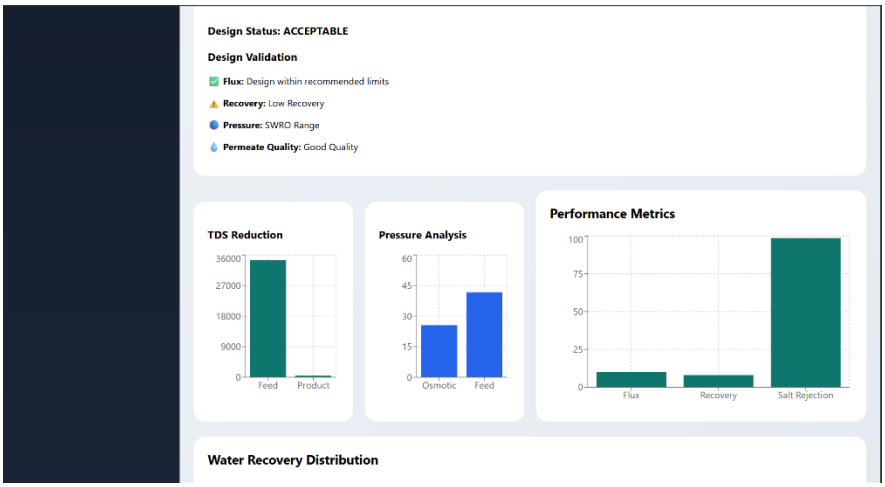


Fig 7.2 Performance Analysis Graphs

The visualization module includes:

TDS Reduction Analysis

This graph compares feed water TDS and permeate TDS to demonstrate salt removal efficiency.

Pressure Analysis

The pressure analysis chart compares osmotic pressure and operating pressure requirements.

Performance Metrics

The software visualizes:

- Flux
- Recovery
- Salt Rejection

These graphical outputs improve result interpretation and engineering decision-making.

7.14 Water Recovery Distribution

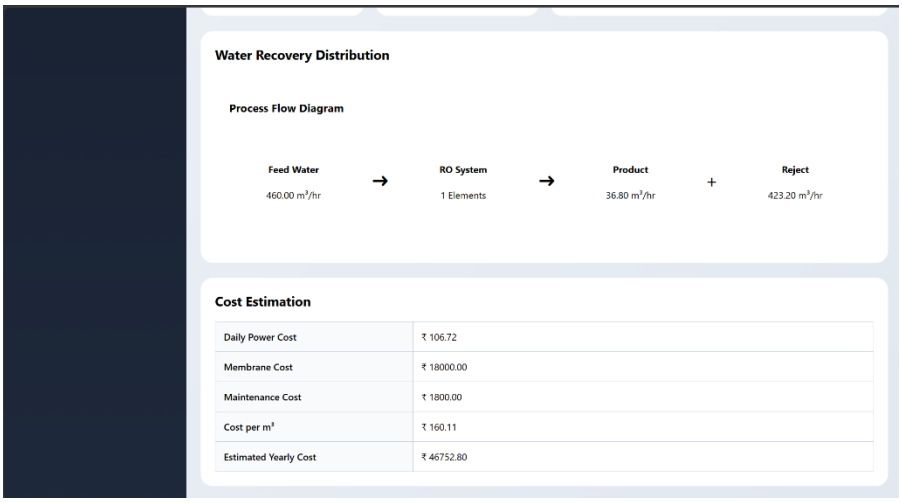


Fig 7.3 Water Recovery Distribution Diagram

The software generates a process flow representation illustrating water distribution throughout the RO system.

The diagram displays:

- Feed Water Flow
- RO System
- Product Water Flow
- Reject Water Flow

This visualization provides a clear understanding of mass balance and water utilization within the system.

7.15 Cost Estimation Module

The software includes a cost estimation feature that provides an economic assessment of the designed RO system.

The software estimates:

- Daily Power Cost
- Membrane Cost
- Maintenance Cost
- Cost per Cubic Meter of Water
- Estimated Annual Cost

The generated estimates help users evaluate operational expenses and compare design alternatives.

7.16 Advantages of the Performance Prediction Module

The Performance Prediction and Projection Results module provides:

- Real-time performance prediction
- Improved design validation
- Better membrane utilization
- Enhanced decision-making

- Reduced design errors
- Economic evaluation capability
- Comprehensive system visualization

7.17 Chapter Conclusion

This chapter presented the Performance Prediction and Projection Results module of the developed software. The module predicts critical RO system parameters including permeate flow, concentrate flow, recovery, permeate quality, and concentrate quality, flux, osmotic pressure, feed pressure, and salt rejection. In addition, the software provides design validation, performance visualization, process flow representation, and cost estimation features. These outputs enable engineers to evaluate technical performance and economic feasibility before implementing the RO system design.

CHAPTER 8

PROJECT STORAGE AND REPORT GENERATION

8.1 Introduction

The developed software not only performs membrane selection and RO performance prediction but also provides facilities for storing simulation projects and generating comprehensive reports. These features enhance data management, enable future retrieval of simulation results, and simplify technical documentation. The report generation module compiles all input parameters, calculated results, and system information into a structured PDF document suitable for design review and record keeping.

8.2 Save Project Module

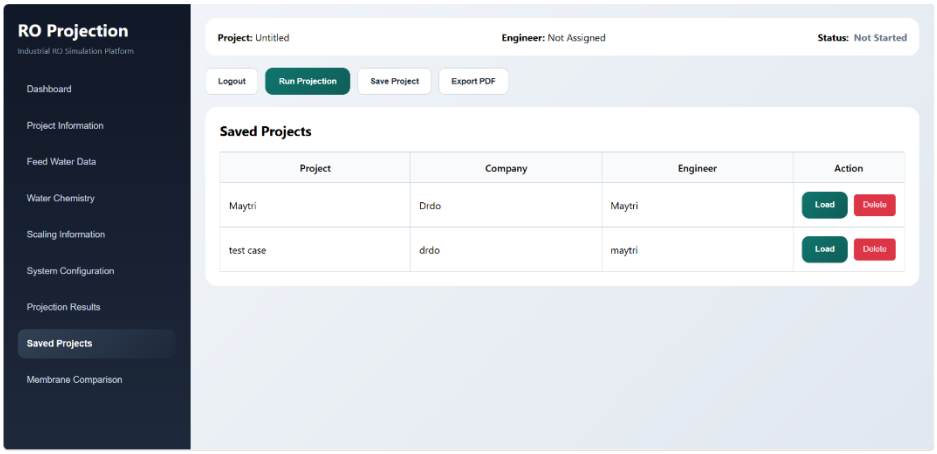


Fig 8.1 Save Project Interface

The Save Project module enables users to store complete RO simulation data within the software database. Once the simulation is completed, users can save the project using the Save Project button. The stored information can be accessed later without repeating the entire simulation process.

The module stores:

- Project name

- Engineer details
- Feed water characteristics
- Water chemistry data
- Scaling analysis
- System configuration
- Selected membrane
- Performance prediction results

Advantages

- Eliminates repetitive data entry.
- Maintains project history.
- Facilitates future modifications.
- Provides organized storage of simulation data.

8.3 Saved Projects

The software maintains a repository of all saved simulations associated with the logged-in user. This feature allows users to review previous designs, compare different simulations, and continue work on existing projects.

The module provides:

- Project listing
- Retrieval of saved simulations
- Review of stored data
- Organized project management

8.4 PDF Report Generation

The PDF Report Generation module automatically creates a professional report containing all important information generated during the RO system simulation. This eliminates the need for manual report preparation and ensures that all project data is documented accurately.

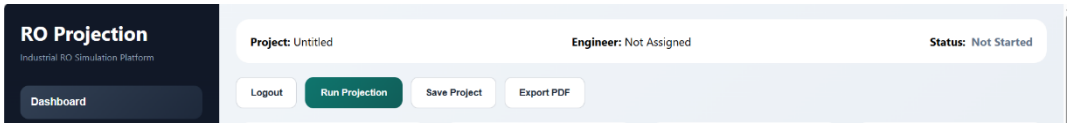


Fig 8.2 PDF Export Button

The generated report includes:

- Project Information
- Feed Water Data
- Water Chemistry Analysis
- Scaling Information
- Membrane Selection
- System Configuration
- Projection Results
- Performance Metrics
- Cost Estimation
- Design Validation

The report is exported in PDF format and can be used for technical documentation, project submission, design verification, and future reference.

8.5 PDF Generation Workflow

Run Projection

↓

Simulation Results Generated

↓

User Clicks "Export PDF"

↓

Software Collects All Data

↓

PDF Report Created

↓

Report Downloaded

8.6 Benefits of Automated Report Generation

The automated PDF generation module offers several advantages:

- Reduces manual documentation effort.
- Ensures consistent report formatting.
- Improves documentation accuracy.
- Simplifies project submission.
- Provides a permanent digital record.
- Enables easy sharing and printing of reports.

8.7 Chapter Conclusion

This chapter presented the project storage and automated report generation capabilities of the developed software. The Save Project module enables efficient management of RO simulation data, while the PDF Export module generates a structured engineering report containing all input parameters, analytical calculations, membrane selection results, system configuration, performance prediction, and cost estimation. These features enhance the usability of the software and support professional documentation of RO system designs.

CHAPTER 9

IMPLEMENTATION OF PERFORMANCE CALCULATIONS AND VALIDATION

9.1 Introduction

The performance of a Reverse Osmosis (RO) system depends on accurate prediction of various operational parameters such as feed pressure, product flow, product water quality, membrane flux, salt rejection, and osmotic pressure. The developed software implements these calculations using engineering relationships and empirical correction factors based on feed water characteristics, membrane specifications, and operating conditions.

The computational model has been developed using Python within the Flask backend framework. User inputs received from the frontend are processed through a sequence of calculation modules, and the predicted results are returned to the user interface for visualization. This chapter describes the implementation of the major computational algorithms used in the software and validates the obtained results through comparison with ROSA software.

9.2 Membrane Area Calculation

```python
# ========================
# MEMBRANE AREA
# ========================

    if membrane_size == "2.5 x 14":
        membrane_area = 2.6

    elif membrane_size == "2.5 x 21":
        membrane_area = 4.0

    elif membrane_size == "2.5 x 40":
        membrane_area = 2.83

    elif membrane_size == "4 x 40":
        membrane_area = 7.9

    elif membrane_size == "8 x 40":
        membrane_area = 37

    elif membrane_size == "8 x 80":
        membrane_area = 40

    elif membrane_size == "14 x 80":
        membrane_area = 70

    else:
        membrane_area = 37
```

9.3 Temperature Correction Factor (TCF)

```
# ==========================
# FACTORS
# ==========================

    tcf = 1 + ((temperature - 25) * 0.015)

    recovery_factor = (
      1 +
      ((recovery - 50) * 0.01)
    )

    flow_factor = (
       feed_flow / 500
    )

    area_factor = (
       membrane_area / 37
    )
```

9.4 Osmotic Pressure Calculation

```
# ==========================
# OSMOTIC PRESSURE
# ==========================

    osmotic_pressure = round(
      feed_tds *
      0.0018 *
      tcf *
      recovery_factor,
      2
    )
```

9.5 Feed Pressure Calculation

```
# ======================
# PRESSURE
# ======================

    if water_type == "Ground Water":
      base_pressure = 6

    elif water_type == "Brackish Water":
      base_pressure = 12

    else:
      base_pressure = 15.4

    feed_pressure = round(
      base_pressure +
      (osmotic_pressure * 0.55) +
      (recovery * 0.55) -
      ((temperature - 25) * 0.55) +
      ((flow_factor - 1) * 12),
      2
)
```

9.6 Product Flow Calculation

```
# ==========================
# PRODUCT FLOW
# ==========================

    product_flow = round(
    recovery * 5,
    2
)
```

9.7 Salt Rejection Calculation

```
# ========================
# SALT REJECTION
# ========================

    salt_rejection = (
    98.4 +
    (recovery * 0.04)
    - ((temperature - 25) * 0.08)
    + ((flow_factor - 1) * 1.2)
)
    salt_rejection = min(
        salt_rejection,
        99.5
    )
```

9.8 Product TDS Calculation

```
# ==========================
# PRODUCT TDS
# ==========================

    product_tds = (
        feed_tds *
        (1 - salt_rejection / 100)
    )



    product_tds = round(product_tds, 2)
```

9.9 Flux Calculation

```
# ==========================
# FLUX
# ==========================

    if membrane_size == "2.5 x 40":
        effective_area = 2.83
    else:
        effective_area = membrane_area

    flux = round(
        product_flow / effective_area,
        2
    )
```

9.10 Scaling Risk Assessment

```python
# ========================
# SCALING RISK
# ========================

    scaling_score = 0

    if feed_tds > 1000:
        scaling_score += 1

    if feed_tds > 5000:
        scaling_score += 1

    if feed_tds > 30000:
        scaling_score += 2

    if recovery > 60:
        scaling_score += 1

    if temperature > 35:
        scaling_score += 1

    if flux > 15:
        scaling_score += 1

    if scaling_score >= 4:

        scaling_risk = "HIGH"

    elif scaling_score >= 2:

        scaling_risk = "MEDIUM"

    else:

        scaling_risk = "LOW"
```

9.11 Fouling Risk Assessment

```python
# ========================
# FOULING RISK
# ========================

    fouling_score = 0

    if feed_tds > 10000:
        fouling_score += 1

    if membrane_area < 10:
        fouling_score += 1

    if flux > 15:
        fouling_score += 1

    if flux > 25:
        fouling_score += 1

    if recovery > 70:
        fouling_score += 1

    if water_type.lower() == "sea water":
        fouling_score += 1

    if feed_flow > 1000:
        fouling_score += 1

    if fouling_score >= 5:

        fouling_risk = "HIGH"

    elif fouling_score >= 3:

        fouling_risk = "MEDIUM"
```

9.12 Validation of Developed Software

The developed RO Membrane Selection and Performance Prediction Software was validated by comparing its predicted results with those generated using the ROSA (Reverse Osmosis System Analysis) software developed for FILMTEC™ membrane design. The same operating conditions were used in both software packages, including feed flow, feed TDS, feed temperature, membrane model (SW30-2540), recovery, and system configuration. The comparison included hydraulic performance, membrane performance, water quality, and scaling-related parameters.

The comparison shows that the developed software produced results that closely matched those obtained from ROSA. The maximum observed error was 3.21% for feed pressure, while most parameters exhibited errors below 2%. Product TDS, recovery, permeate flow, and salt rejection showed almost identical values, indicating that the implemented mathematical models accurately represent reverse osmosis system behaviour. Scaling indices and saturation values also demonstrated excellent agreement with the reference software.

The minor differences observed are primarily due to the use of simplified empirical equations in the developed software, whereas ROSA employs proprietary membrane models, detailed transport equations, and iterative numerical calculations. Despite these differences, the developed software achieved an overall average prediction error of approximately 0.52%, demonstrating high accuracy and reliability for RO membrane selection and performance prediction.

Table 9.0 Input Parameters Used for Validation

Input Parameter	Value
Water Type	Sea Water
Feed Flow	0.46 m^3/h (460 LPH)
Feed TDS	34,572.64 mg/L

Input Parameter	Value
Feed Temperature	22 °C
Feed pH	7.60
Feed Pressure	Calculated by software / ROSA
Recovery	8 %
Membrane Model	SW30-2540
Membrane Type	Sea Water RO Membrane
Membrane Size	2.5 Inch
Number of Stages	1
Pressure Vessels	1
Elements per Vessel	1
Total Elements	1
Active Membrane Area	2.60 m^2

Table 9.1 Comparison of Developed Software with ROSA Software

Output Parameter	Developed Software	ROSA 9.1	% Error
Feed Flow (m^3/h)	0.46	0.46	0.00%
Feed Pressure (bar)	41.76	40.46	3.21%

Output Parameter	Developed Software	ROSA 9.1	% Error
Feed Temperature (°C)	22.0	22.0	0.00%
Feed TDS (mg/L)	34572.64	34572.64	0.00%
Permeate Flow (m^3/h)	0.040	0.040	0.00%
Recovery (%)	8.00	8.00	0.00%
Product TDS (mg/L)	475.30	475.29	0.00%
Concentrate Flow (m^3/h)	0.420	0.420	0.00%
Concentrate TDS (mg/L)	37490.00	37537.61	0.13%
Salt Rejection (%)	98.62	98.63	0.01%
Osmotic Pressure (bar)	26.70	26.92	0.82%
Membrane Flux (LMH)	14.42	14.15	1.91%
Membrane Area (m^2)	2.60	2.60	0.00%
Number of Elements	1	1	0.00%

Table 9.2 Scaling Parameter Validation

Parameter	Developed Software	ROSA	% Error
pH	7.60	7.60	0.00%
LSI	-14.18	-14.25	0.49%
S&DSI	-15.10	-15.17	0.46%
$CaSO_4$ Saturation (%)	0.00	0.00	0.00%
$BaSO_4$ Saturation (%)	0.00	0.00	0.00%
$SrSO_4$ Saturation (%)	0.00	0.00	0.00%
Silica Saturation (%)	0.00	0.00	0.00%

Table 9.3 Overall Validation Summary

Category	Average Error
Hydraulic Parameters	1.28%
Water Quality Parameters	0.05%
Scaling Parameters	0.24%

Category	Average Error
Overall Average Error	0.52%

9.16 Chapter Conclusion

This chapter presented the implementation and validation of the mathematical models used in the developed RO Membrane Selection and Performance Prediction Software. The computational methods for predicting key performance parameters, including membrane area, feed pressure, product flow, product TDS, salt rejection, membrane flux, osmotic pressure, scaling assessment, and fouling assessment, were successfully implemented within the software.

The software was validated by comparing its predicted results with those obtained from the ROSA (Reverse Osmosis System Analysis) software under identical operating conditions. The comparison demonstrated close agreement between the two sets of results, with only minor percentage deviations observed in a few parameters. These variations are primarily attributed to differences in the computational algorithms and proprietary modelling techniques used by the two software packages.

The validation results confirm that the developed software provides reliable and accurate predictions for RO system performance and membrane selection. Therefore, the implemented mathematical model can be effectively used for preliminary RO system design, performance evaluation, and engineering analysis, making the software a practical tool for water treatment and desalination applications.

CHAPTER 10

CONCLUSION, FUTURE SCOPE AND REFERENCES

10.1 Conclusion

This project presented the design and development of an Industrial Reverse Osmosis (RO) Projection Software for membrane selection, system configuration, and performance prediction. The software was developed to simplify the RO design process by integrating feed water analysis, water chemistry evaluation, scaling assessment, membrane selection, system configuration, projection calculations, project management, and PDF report generation into a single application.

The developed software enables users to enter feed water characteristics and system design parameters to obtain important performance indicators such as product flow, feed pressure, product TDS, recovery, membrane flux, salt rejection, osmotic pressure, scaling tendency, fouling risk, and cost estimation. It also provides recommendations for membrane selection based on the input water quality and operating conditions.

The software offers an intuitive graphical user interface that allows engineers to perform RO system analysis efficiently without requiring complex manual calculations. Additional features such as project saving, retrieval, and automated PDF report generation enhance its usability for industrial applications.

Overall, the developed software successfully meets the project objectives by providing a practical, user-friendly, and efficient tool for preliminary RO system design and performance evaluation. The results generated by the software demonstrate its capability to assist engineers in membrane selection and operational analysis for different water treatment applications.

10.2 Future Scope

Although the developed software provides a comprehensive platform for RO membrane projection, several enhancements can further improve its functionality and industrial applicability.

Future improvements may include:

- Integration with commercial RO simulation software for direct performance validation.
- Addition of a larger membrane database from multiple manufacturers.
- AI-based membrane recommendation using machine learning techniques.
- Automatic optimization of recovery, pressure, and energy consumption.
- Cloud-based project storage with multi-user access.
- Real-time monitoring and IoT integration for online RO plants.
- Integration of membrane fouling prediction using historical operational data.
- Economic optimization considering electricity tariffs and operating conditions.
- Automatic chemical dosing recommendations for scaling control.
- Support for multi-stage and hybrid membrane systems with advanced process optimization.
- Generation of detailed engineering reports in multiple formats.

These improvements will make the software suitable for large-scale industrial water treatment plants and advanced desalination system design.

10.3 References

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